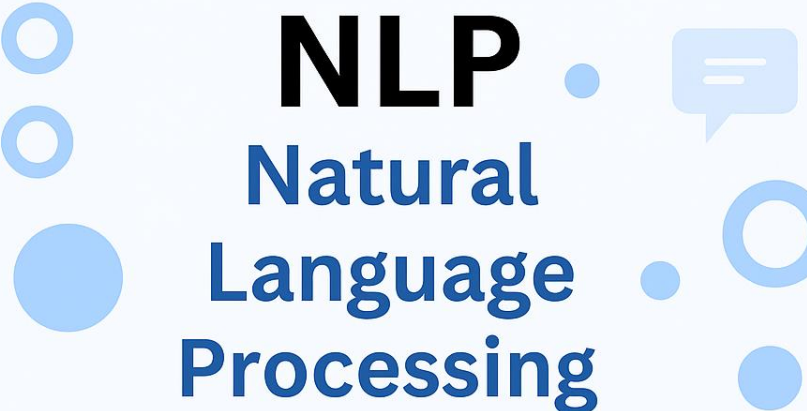
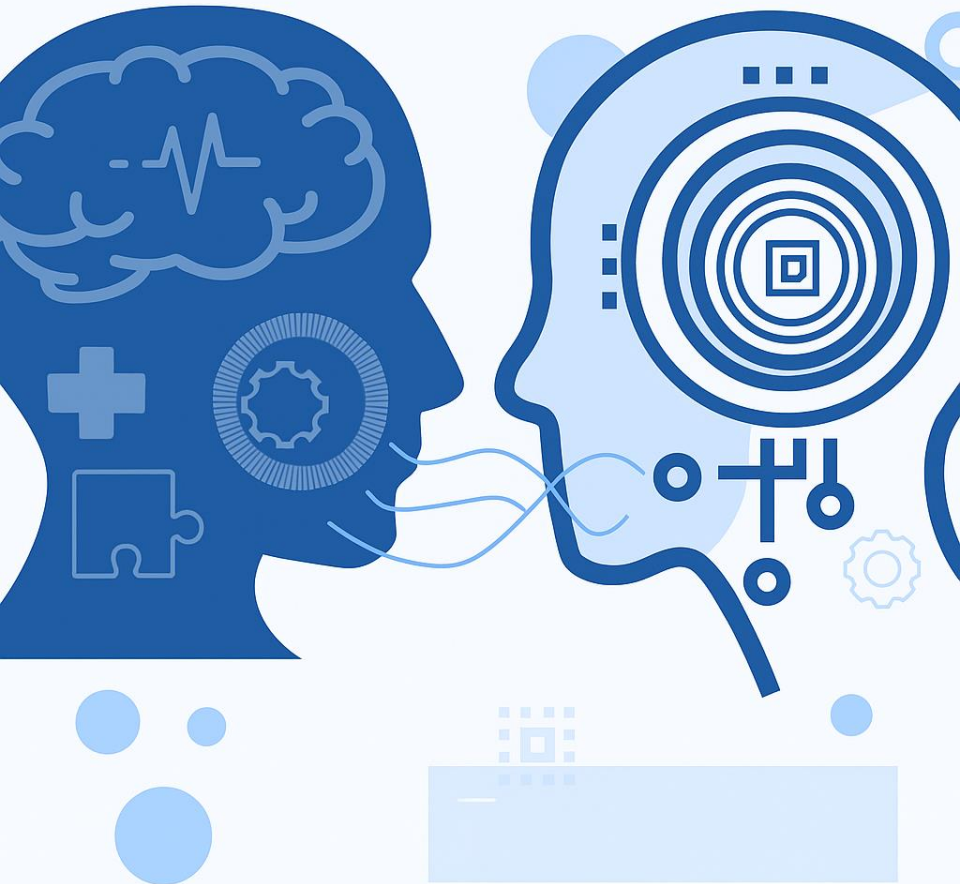




NLP

Natural
Language
Processing



NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 5

LECTURE 1

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TYPES OF SYNTACTIC AMBIGUITY

- Attachment Ambiguity
- Coordination Ambiguity
- Complementation Ambiguity
- Scope Ambiguity
- Elliptical / Anaphoric Ambiguity
- Garden-Path Ambiguity

ATTACHMENT AMBIGUITY

Occurs when a phrase (often a prepositional phrase or adverbial phrase) can attach to different parts of the sentence.

Prepositional Phrase (PP) Attachment

Example: “***I saw the man with a telescope.***”

Parse 1: (I saw) [the man with a telescope] → ***The man had the telescope.***

Parse 2: (I saw) [the man] (with a telescope) → ***I used the telescope to see him.***

Adverb Attachment

Example: “***He said she left yesterday.***”

Parse 1: ***He said it yesterday.***

Parse 2: ***She left yesterday.***

Reason: The parser must decide whether the modifier attaches to the verb phrase (VP) or noun phrase (NP).

COORDINATION AMBIGUITY

Occurs with conjunctions like **and**, **or**, **but**, where it's unclear which parts are conjoined.

Example: ***“I saw the man and the woman with a telescope.”***

Parse 1: (I saw) [the man and the woman] (with a telescope).

Parse 2: (I saw) [the man] and [the woman with a telescope].

Example: ***“Old men and women.”***

Parse 1: Old [men and women] → Both are old.

Parse 2: [Old men] and women → Only men are old.

Reason: Conjunction can coordinate different categories or phrases, leading to multiple parses.

COMPLEMENTATION AMBIGUITY

Occurs when it is unclear how a complement clause or infinitive phrase attaches.

Example: “***John knows that Mary is easy to please.***”

Parse 1: Mary is easy to be pleased (object interpretation).

Parse 2: Mary is easy at pleasing others (subject interpretation).

Example: “***Visiting relatives can be annoying.***”

Parse 1: The act of visiting relatives is annoying.

Parse 2: Relatives who are visiting are annoying.

Reason: Complement clauses (that-clauses, infinitives, gerunds) can take on multiple syntactic roles.

SCOPE AMBIGUITY

Occurs when quantifiers, negation, or modals create different possible scopes.

Example: “***Every student read a book.***”

Parse 1: There exists one book that every student read ($\exists > \forall$).

Parse 2: Each student read possibly different books ($\forall > \exists$).

Example: “***John didn’t eat every cake.***”

Parse 1: John ate none of the cakes.

Parse 2: John ate some, but not all cakes.

Reason: Multiple logical representations from the same syntactic structure.

ELLIPTICAL / ANAPHORIC AMBIGUITY

Occurs when ellipsis or reference leaves the structure under-specified.

Example: “***John likes his dog more than Bill.***”

Parse 1: John likes his dog more than Bill likes John’s dog.

Parse 2: John likes his dog more than he likes Bill.

Example: “***I do too.***” (*ellipsis*)

Meaning depends on what is being agreed to (I like pizza too, I run every morning too).

Reason: Pronouns, ellipsis, and anaphora depend on external context, creating syntactic ambiguity.

GARDEN-PATH AMBIGUITY

Occurs when the parser is led to one interpretation, but the continuation of the sentence forces a re-analysis.

Example: “***The horse raced past the barn fell.***”

At first: (The horse raced past the barn) is complete.

Actual: The horse [that was raced past the barn] fell.

Example: “***The old man the boats.***”

Looks like “The old man” is subject.

Actual parse: “The old [people] man (operate) the boats.”

Reason: Reduced relative clauses and unusual structures mislead the parser.

CONTEXT-FREE GRAMMAR (CFG)

- A Context-Free Grammar (CFG) is a formal system used to describe the syntax (structure) of a language.
- It is widely used in linguistics, compilers, and natural language processing (NLP) to specify how sentences (or strings) can be formed from smaller units.
- In computer science, CFGs are fundamental to defining the syntax of programming languages, and they are used by compilers to parse source code
- Invented by: Noam Chomsky (1956).
- “Context-Free” means rules apply regardless of the surrounding symbols (i.e., independent of context), i.e., the substitution of a symbol in a production rule is not dependent on the symbols surrounding it.

COMPONENTS OF CFG

- **V (Variables/Non-terminals)**: Symbols that can be further expanded. E.g., **S** (sentence), **NP** (noun phrase), **VP** (verb phrase).
- **Σ (Alphabet/Terminals)**: The actual words or tokens of the language. E.g., {dog, barks, eats, pizza}.
- **R (Production Rules)**: Rewrite rules that define how non-terminals can be replaced. Each rule has the form:
$$A \rightarrow \alpha$$

where A is a non-terminal, and α is a sequence of terminals and/or non-terminals.
- **S (Start Symbol)**: A special non-terminal that represents a complete sentence.

PRODUCTION RULES

NP → *Det Nominal*
NP → *ProperNoun*
Nominal → *Noun | Nominal Noun*

An ***NP*** (or noun phrase) can be composed of either a ***ProperNoun*** or a ***determiner (Det)*** followed by a ***Nominal***; a ***Nominal*** in turn can consist of one or more ***Nouns***.

Context-free rules can be hierarchically embedded, so we can combine the previous rules with others, like the following, that express facts about the lexicon:

Det → *a*
Det → *the*
Noun → *flight*

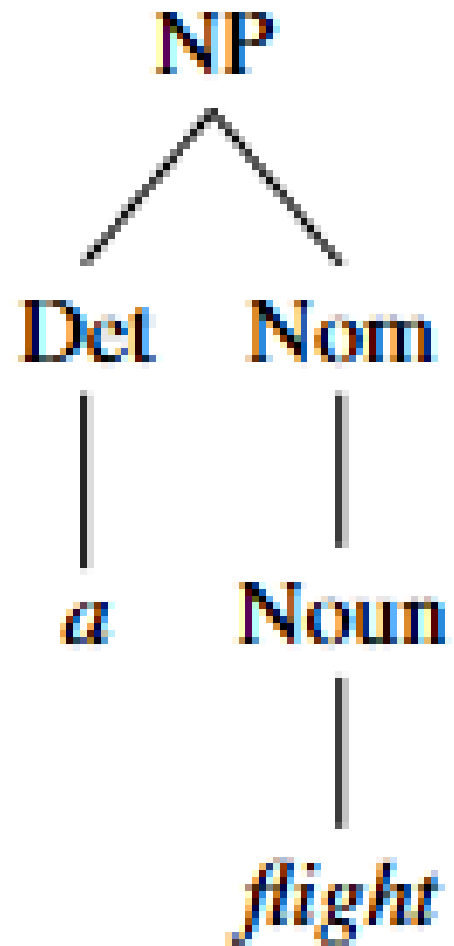
PRODUCTION RULES

A CFG can be thought of in two ways:

- as a device for generating sentences and as a device for assigning a structure to a given sentence.
- Viewing a CFG as a generator, we can read the $\rightarrow$ arrow as “rewrite the symbol on the left with the string of symbols on the right”

***So starting from the symbol: NP
we can use our first rule to rewrite NP as: Det Nominal
and then rewrite Nominal as: Noun
and finally rewrite these parts-of-speech as: a flight***

PRODUCTION RULES



EXAMPLE

$S \rightarrow NP VP$ I prefer a morning flight

A verb phrase in English consists of a verb followed by assorted other things; for example, one kind of verb phrase consists of a verb followed by a noun phrase:

$VP \rightarrow Verb NP$ prefer a morning flight

Or the verb may be followed by a noun phrase and a prepositional phrase:

$VP \rightarrow Verb NP PP$ leave Boston in the morning

Or the verb phrase may have a verb followed by a prepositional phrase alone:

$VP \rightarrow Verb PP$ leaving on Thursday

A prepositional phrase generally has a preposition followed by a noun phrase. For example, a common type of prepositional phrase in the ATIS corpus is used to indicate location or direction:

$PP \rightarrow Preposition NP$ from Los Angeles

EXAMPLE

Noun → *flights* | *flight* | *breeze* | *trip* | *morning*

Verb → *is* | *prefer* | *like* | *need* | *want* | *fly* | *do*

Adjective → *cheapest* | *non-stop* | *first* | *latest*
| *other* | *direct*

Pronoun → *me* | *I* | *you* | *it*

Proper-Noun → *Alaska* | *Baltimore* | *Los Angeles*
| *Chicago* | *United* | *American*

Determiner → *the* | *a* | *an* | *this* | *these* | *that*

Preposition → *from* | *to* | *on* | *near* | *in*

Conjunction → *and* | *or* | *but*

EXAMPLE

Grammar Rules	Examples
$S \rightarrow NP VP$	I + want a morning flight
$NP \rightarrow$ <i>Pronoun</i> <i>Proper-Noun</i> <i>Det Nominal</i>	I Los Angeles a + flight
$Nominal \rightarrow$ <i>Nominal Noun</i> <i>Noun</i>	morning + flight flights
$VP \rightarrow$ <i>Verb</i> <i>Verb NP</i> <i>Verb NP PP</i> <i>Verb PP</i>	do want + a flight leave + Boston + in the morning leaving + on Thursday
$PP \rightarrow$ <i>Preposition NP</i>	from + Los Angeles

EXAMPLE

